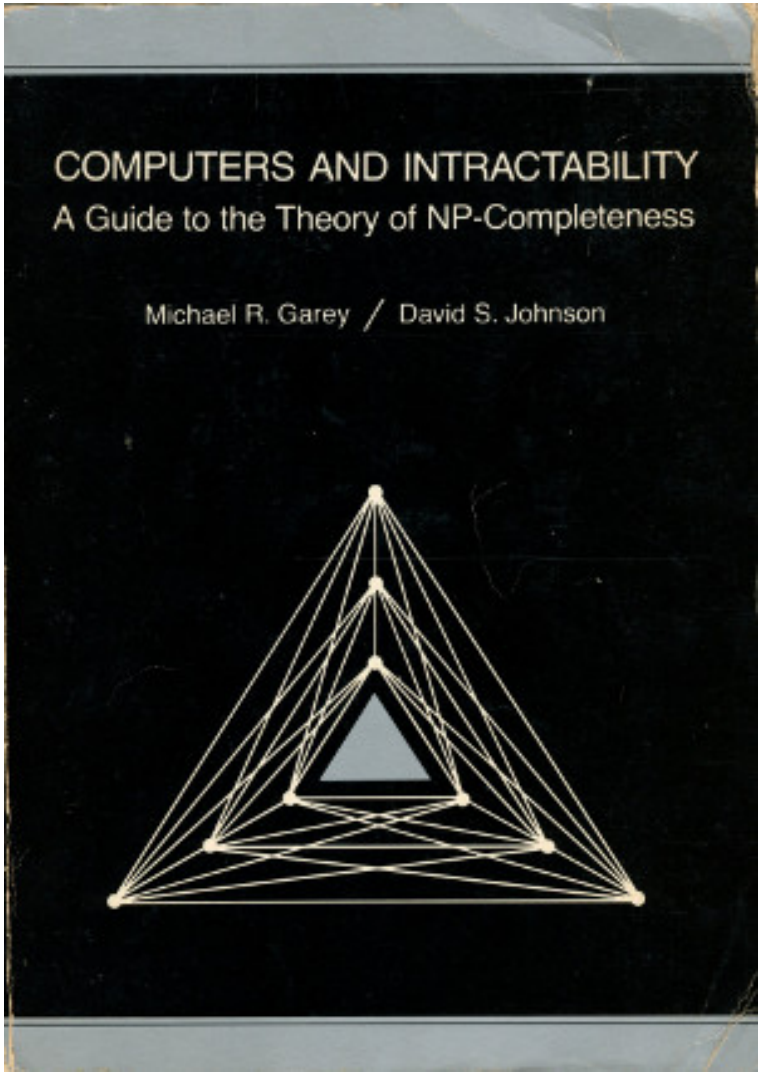
The complexity of hard graph problems forty years later

Celina Miraglia Herrera de Figueiredo



Workshop ParGO 20+50, Universidade Federal do Ceará, agosto 2019

The Guide – Computers and Intractability



"Despite that 23 years have passed since its publication, I consider Garey and Johnson the single most important book on my office bookshelf. Every computer scientist should have this book on their shelves as well. NP-completeness is the single most important concept to come out of theoretical computer science and no book covers it as well as Garey and Johnson."

Lance Fortnow, "Great Books: Computers and Intractability: A Guide to the Theory of NP-Completeness"

NP-completo: simboliza o abismo da intratabilidade inerente para resolver problemas maiores e mais complexos

Variedade ampla de problemas frequentes: matemática, computação, pesquisa operacional

- Capítulos 1–5: teoria básica
- Capítulos 6–7: aproximação, hierarquia de classes de complexidade
- Apêndice: metade do livro! Lista bem organizada de problemas

Ongoing Guide – Os 12 problemas atualizados em 2005

Problem Name	Source	Status	Covered in
GRAPH ISOMORPHISM	[G&J]	Open	–
SUBGRAPH HOMEOMORPHISM (FOR A FIXED GRAPH H)	[G&J]	P	[Col 19, 1987]
GRAPH GENUS	[G&J]	NPC	[Col 21, 1988]
CHORDAL GRAPH COMPLETION	[G&J]	NPC	[Col 1, 1981]
CHROMATIC INDEX	[G&J]	NPC	[Col 1, 1981]
PARTIAL ORDER DIMENSION	[G&J]	NPC	[Col 1, 1981]
PRECEDENCE CONSTRAINED 3-PROCESSOR SCHEDULING	[G&J]	Open	–
LINEAR PROGRAMMING	[G&J]	P	[Col 1, 1981]
TOTAL UNIMODULARITY	[G&J]	P	[Col 1, 1981]
SPANNING TREE PARITY PROBLEM	[G&J]	P	[Col 1, 1981]
COMPOSITE NUMBER	[G&J]	P	This Column
MINIMUM LENGTH TRIANGULATION	[G&J]	Open	–
IMPERFECT GRAPH	[Col 1, 1981]	P	This Column
GRAPH THICKNESS	[Col 2, 1982]	NPC	[Col 5, 1982]
EVEN COVER (MINIMUM WEIGHT CODEWORD)	[Col 3, 1982]	NPC	This Column
"UNRESTRICTED" TWO-LAYER CHANNEL ROUTING	[Col 5, 1982]	Open	–
GRACEFUL GRAPH	[Col 6, 1983]	Open	–
ANDREEV'S PROBLEM	[Col 17, 1986]	Open	–
SHORTEST VECTOR IN A LATTICE	[Col 18, 1986]	"NPC"	This Column

Ongoing Guide – Graph Restrictions and Their Effect

Graph Class	Member		IndSet		Clique		CliPar		ChrNum	
Trees/Forests	P	[T]	P	[GJ]	P	[T]	P	[GJ]	P	[T]
Almost Trees (k)	P		P	[24]	P	[T]	P?		P?	
Partial k-Trees	P	[2]	P	[1]	P	[T]	P?		P	[1]
Bandwidth-k	P	[68]	P	[64]	P	[T]	P?		P	[64]
Degree-k	P	[T]	N	[GJ]	P	[T]	N	[GJ]	N	[GJ]
Planar	P	[GJ]	N	[GJ]	P	[T]	N	[10]	N	[GJ]
Series Parallel	P	[79]	P	[75]	P	[T]	P?		P	[74]
Outerplanar	P		P	[6]	P	[T]	P	[6]	P	[67]
Halin	P		P	[6]	P	[T]	P	[6]	P	[74]
k-Outerplanar	P		P	[6]	P	[T]	P	[6]	P	[6]
Grid	P		P	[GJ]	P	[T]	P	[GJ]	P	[T]
$K_{3,3}$-Free	P	[4]	N	[GJ]	P	[T]	N	[10]	N	[GJ]
Thickness-k	N	[60]	N	[GJ]	P	[T]	N	[10]	N	[GJ]
Genus-k	P	[34]	N	[GJ]	P	[T]	N	[10]	N	[GJ]
Perfect	O!		P	[42]	P	[42]	P	[42]	P	[42]
Chordal	P	[76]	P	[40]	P	[40]	P	[40]	P	[40]
Split	P	[40]	P	[40]	P	[40]	P	[40]	P	[40]
Strongly Chordal	P	[31]	P	[40]	P	[40]	P	[40]	P	[40]
Comparability	P	[40]	P	[40]	P	[40]	P	[40]	P	[40]
Bipartite	P	[T]	P	[GJ]	P	[T]	P	[GJ]	P	[T]
Permutation	P	[40]	P	[40]	P	[40]	P	[40]	P	[40]
Cographs	P	[T]	P	[40]	P	[40]	P	[40]	P	[40]
Undirected Path	P	[39]	P	[40]	P	[40]	P	[40]	P	[40]
Directed Path	P	[38]	P	[40]	P	[40]	P	[40]	P	[40]
Interval	P	[17]	P	[44]	P	[44]	P	[44]	P	[44]
Circular Arc	P	[78]	P	[44]	P	[50]	P	[44]	N	[36]
Circle	P	[71]	P	[GJ]	P	[50]	O?		N	[36]
Proper Circ. Arc	P	[77]	P	[44]	P	[50]	P	[44]	P	[66]
Edge (or Line)	P	[47]	P	[GJ]	P	[T]	N	[GJ]	N	[49]
Claw-Free	P	[T]	P	[63]	O?		N	[GJ]	N	[49]

ChrInd		HamCir		DomSet		MaxCut		StTree		GraIso	
P	[GJ]	P	[T]	P	[GJ]	P	[GJ]	P	[T]	P	[GJ]
P?		P?		P	[45]	P?		P?		P?	
O?		P	[3]	P	[3]	P?		P?		O?	
P?		P?		P	[64]	P	[64]	P?		P	[58]
N	[49]	N	[GJ]	N	[GJ]	N	[GJ]	N	[GJ]	P	[58]
O		N	[GJ]	N	[GJ]	P	[GJ]	N	[35]	P	[GJ]
P	[74]	P	[74]	P	[54]	P	[GJ]	P	[82]	P	[GJ]
P	[67]	P	[T]	P	[6]	P	[GJ]	P	[81]	P	[GJ]
P	[74]	P	[T]	P	[6]	P	[GJ]	P?		P	[GJ]
O?		P	[6]	P	[6]	P	[GJ]	P?		P	[GJ]
P	[GJ]	N	[51]	N	[55]	P	[T]	N	[35]	P	[GJ]
O?		N	[GJ]	N	[GJ]	P	[5]	N	[GJ]	O?	
N	[49]	N	[GJ]	N	[GJ]	N	[7]	N	[GJ]	O?	
O?		N	[GJ]	N	[GJ]	O?		N	[GJ]	P	[61]
O?		N	[1]	N	[14]	O?		N	[GJ]	I	[GJ]
O?		N	[22]	N	[14]	O?		N	[83]	I	[GJ]
O?		N	[22]	N	[19]	O?		N	[83]	I	[15]
O?		O?		P	[32]	O?		P	[83]	O?	
O?		N	[1]	N	[28]	O?		N	[GJ]	I	[GJ]
P	[GJ]	N	[1]	N	[28]	P	[T]	N	[GJ]	I	[GJ]
O?		O		P	[33]	O?		P	[23]	P	[21]
O?		P	[25]	P	[33]	O?		P	[23]	P	[25]
O?		O?		N	[16]	O?		O?		I	[GJ]
O?		O?		P	[16]	O?		P	[83]	O?	
O?		P	[53]	P	[16]	O?		P	[83]	P	[57]
O?		O?		P	[13]	O?		P	[83]	O?	
O?		P	[12]	O?		O?		P	[70]	O?	
O?		P	[12]	P	[13]	O?		P	[83]	O?	
O?		N	[11]	N	[GJ]	O?		N	[70]	I	[15]
O?		N	[11]	N	[GJ]	O?		N	[70]	I	[15]

MaxCut is hard when restricted to geometric intersection model graph classes

Celina Miraglia Herrera de Figueiredo



CLAIO 2022

Based on

Maximum cut on interval graphs of interval count four is NP-complete
with Alexsander A. de Melo, Fabiano S. Oliveira, Ana Silva
arxiv.org/abs/2012.09804



MaxCut on Permutation Graphs is NP-complete
with Alexsander A. de Melo, Fabiano S. Oliveira, Ana Silva
arxiv.org/abs/2202.13955

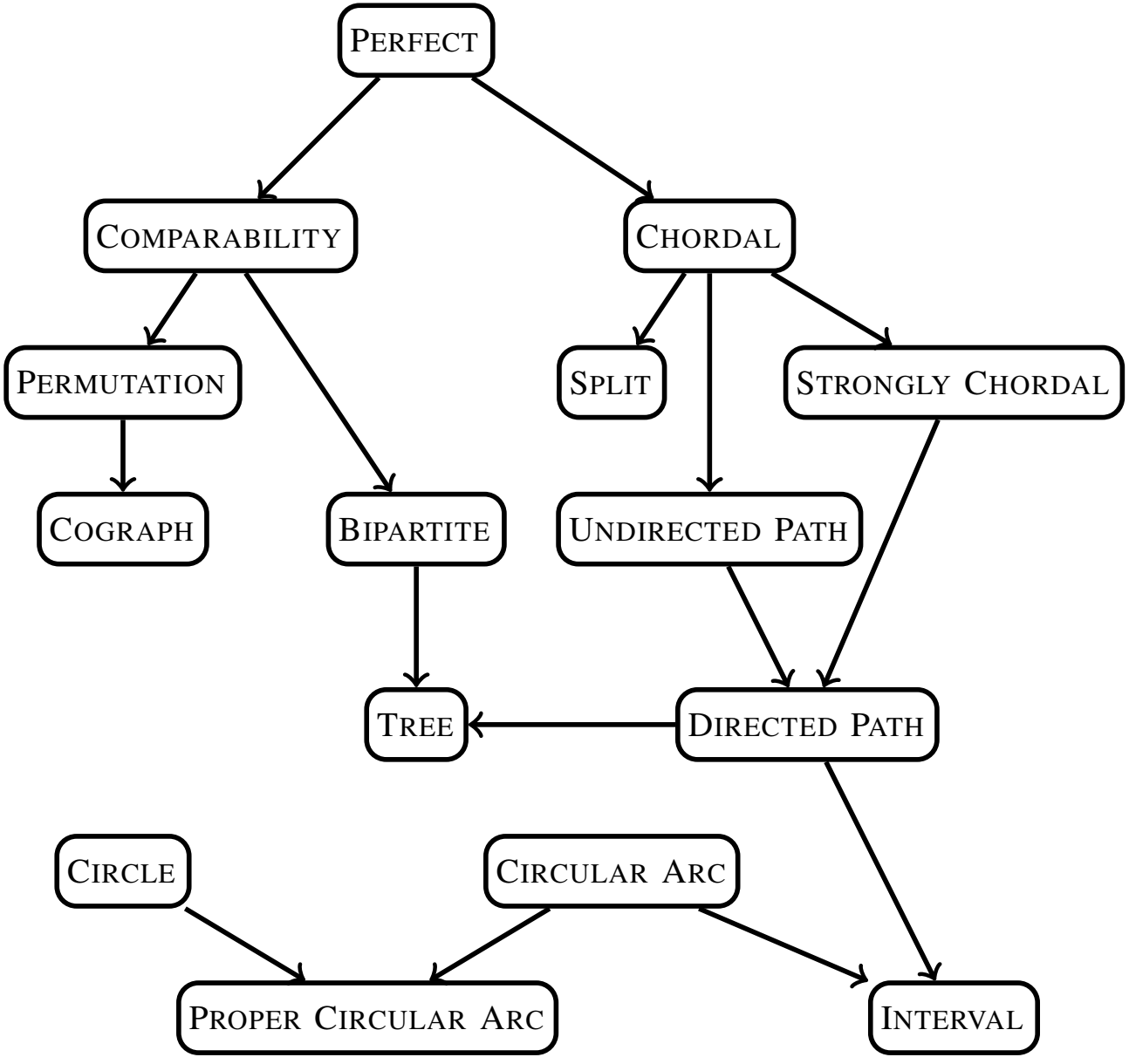


The updated NP-Completeness Column: An Ongoing Guide table 35 years later

Graph Class	Member		IndSet		Clique		CliPar		ChrNum		ChrInd		HamCir		DomSet		MaxCut		StTree		GraphIso	
Trees/Forests	P	[T]	P	[GJ]	P	[T]	P	[GJ]	P	[T]	P	[GJ]	P	[T]	P	[GJ]	P	[GJ]	P	[T]	P	[GJ]
Almost Trees (k)	P	[OG]	P	[OG]	P	[T]	P	[105]	P	[5]	P	[17]	P	[5]	P	[5]	P	[20]	P	[76]	P	[17]
Partial k-trees	P	[OG]	P	[5]	P	[T]	P	[105]	P	[5]	P	[17]	P	[5]	P	[5]	P	[20]	P	[76]	P	[17]
Bandwidth-k	P	[OG]	P	[OG]	P	[T]	P	[105]	P	[5]	P	[17]	P	[5]	P	[5]	P	[OG]	P	[76]	P	[OG]
Degree-k	P	[T]	N	[GJ]	P	[T]	N	[29]	N	[GJ]	N	[OG]	N	[GJ]	N	[GJ]	N	[GJ]	N	[GJ]	P	[OG]
Planar	P	[GJ]	N	[GJ]	P	[T]	N	[78]	N	[GJ]	O		N	[GJ]	N	[GJ]	P	[GJ]	N	[OG]	P	[GJ]
Series Parallel	P	[OG]	P	[OG]	P	[T]	P	[105]	P	[5]	P	[17]	P	[5]	P	[OG]	P	[GJ]	P	[OG]	P	[GJ]
Outerplanar	P	[OG]	P	[OG]	P	[T]	P	[OG]	P	[OG]	P	[OG]	P	[T]	P	[OG]	P	[GJ]	P	[OG]	P	[GJ]
Halin	P	[OG]	P	[OG]	P	[T]	P	[OG]	P	[5]	P	[17]	P	[T]	P	[OG]	P	[GJ]	P	[118]	P	[GJ]
k-Outerplanar	P	[OG]	P	[OG]	P	[T]	P	[OG]	P	[5]	P	[17]	P	[OG]	P	[OG]	P	[GJ]	P	[76]	P	[GJ]
Grid	P	[OG]	P	[GJ]	P	[T]	P	[GJ]	P	[T]	P	[GJ]	N	[OG]	N	[32]	P	[T]	N	[OG]	P	[GJ]
$K_{3,3}$-Free*	P	[OG]	N	[GJ]	P	[T]	N	[78]	N	[GJ]	O?		N	[GJ]	N	[GJ]	P	[OG]	N	[GJ]	P	[40]
Thickness-k	N	[OG]	N	[GJ]	P	[T]	N	[78]	N	[GJ]	N	[OG]	N	[GJ]	N	[GJ]	N	[119]	N	[GJ]	I	[RJ]
Genus-k	P	[OG]	N	[GJ]	P	[T]	N	[78]	N	[GJ]	O?		N	[GJ]	N	[GJ]	O?		N	[GJ]	P	[OG]
Perfect	P	[34]	P	[OG]	P	[OG]	P	[OG]	P	[OG]	N	[28]	N	[OG]	N	[OG]	N	[20]	N	[GJ]	I	[84]
Chordal	P	[OG]	P	[OG]	P	[OG]	P	[OG]	P	[OG]	O?		N	[93]	N	[OG]	N	[20]	N	[OG]	I	[84]
Split	P	[OG]	P	[OG]	P	[OG]	P	[OG]	P	[OG]	O?		N	[93]	N	[OG]	N	[20]	N	[OG]	I	[108]
Strongly Chordal	P	[OG]	P	[OG]	P	[OG]	P	[OG]	P	[OG]	O?		N	[93]	P	[OG]	N	[109]	P	[OG]	I	[111]
Comparability	P	[OG]	P	[OG]	P	[OG]	P	[OG]	P	[OG]	N	[28]	N	[OG]	N	[94]	N	[102]	N	[GJ]	I	[22]
Bipartite	P	[T]	P	[GJ]	P	[T]	P	[GJ]	P	[T]	P	[GJ]	N	[OG]	N	[94]	P	[T]	N	[GJ]	I	[22]
Permutation	P	[OG]	P	[OG]	P	[OG]	P	[OG]	P	[OG]	O?		P	[44]	P	[OG]	N	[120]	P	[OG]	P	[OG]
Cographs	P	[T]	P	[OG]	P	[OG]	P	[OG]	P	[OG]	O?		P	[OG]	P	[OG]	P	[20]	P	[OG]	P	[OG]
Undirected Path	P	[OG]	P	[OG]	P	[OG]	P	[OG]	P	[OG]	O?		N	[13]	N	[OG]	N	[20]	N	[RJ]	I	[22]
Directed Path	P	[OG]	P	[OG]	P	[OG]	P	[OG]	P	[OG]	O?		N	[99]	P	[OG]	N	[1]	P	[OG]	P	[7]
Interval	P	[OG]	P	[OG]	P	[OG]	P	[OG]	P	[OG]	O?		P	[OG]	P	[OG]	N	[1]	P	[OG]	P	[OG]
Circular Arc	P	[OG]	P	[OG]	P	[OG]	P	[OG]	N	[OG]	O?		P	[106]	P	[OG]	N	[1]	P	[11]	P	[80]
Circle	P	[OG]	P	[GJ]	P	[OG]	N	[73]	N	[OG]	O?		N	[39]	N	[71]	N	[26]	P	[OG]	P	[68]
Proper Circ. Arc	P	[OG]	P	[OG]	P	[OG]	P	[OG]	P	[OG]	O?		P	[OG]	P	[OG]	O?		P	[11]	P	[82]
Edge (or Line)	P	[OG]	P	[GJ]	P	[T]	N	[95]	N	[OG]	N	[28]	N	[OG]	N	[GJ]	P	[59]	N	[19]	I	[OG]
Claw-Free	P	[T]	P	[OG]	N	[103]	N	[85]	N	[OG]	N	[28]	N	[OG]	N	[GJ]	N	[20]	N	[19]	I	[OG]

www.cos.ufrj.br/~celina/ftp/j/RJ-current.pdf

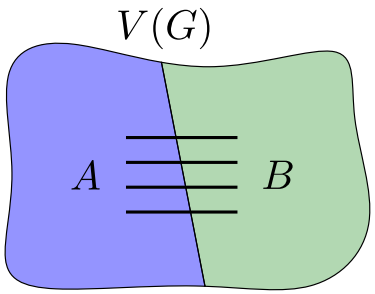
Containment relations for classes



MaxCut problem

Given a graph G and $k \in \mathbb{Z}_0^+$, MaxCut asks whether

$$\mathrm{mc}(G) = \max_{[A,B]} |E_G(A, B)| \geq k.$$



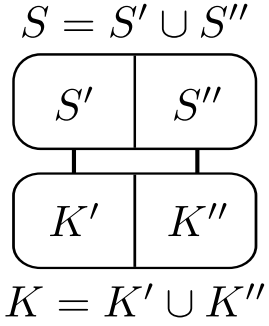
Classical NP-complete problem

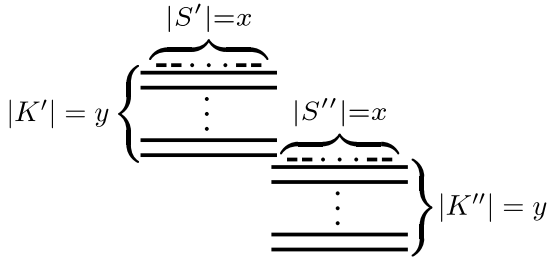
(Garey, Johnson, Stockmeyer, 1976).

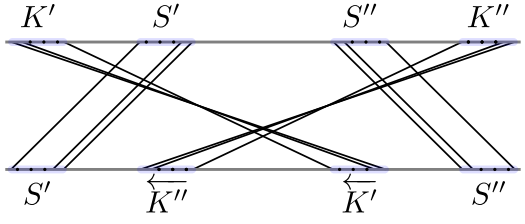
The key gadget to the NP-completeness

An (x, y)-grained gadget is a split graph $H\langle K, S\rangle$, such that

- $S = S' \cup S''$, $|S'| = |S''| = x$;
- $K = K' \cup K''$, $|K'| = |K''| = y$;
- $N_H(K') = K \cup S'$;
- $N_H(K'') = K \cup S''$.





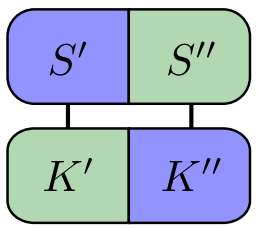


The key property of grained gadgets

Let G be a graph and $[A, B]$ be a maximum cut of G.

If G respects the structure of an (x, y)-grained gadget H, then, for suitable x and y,

- either H is A-partitioned by $[A, B]$;
- or H is B-partitioned by $[A, B]$.

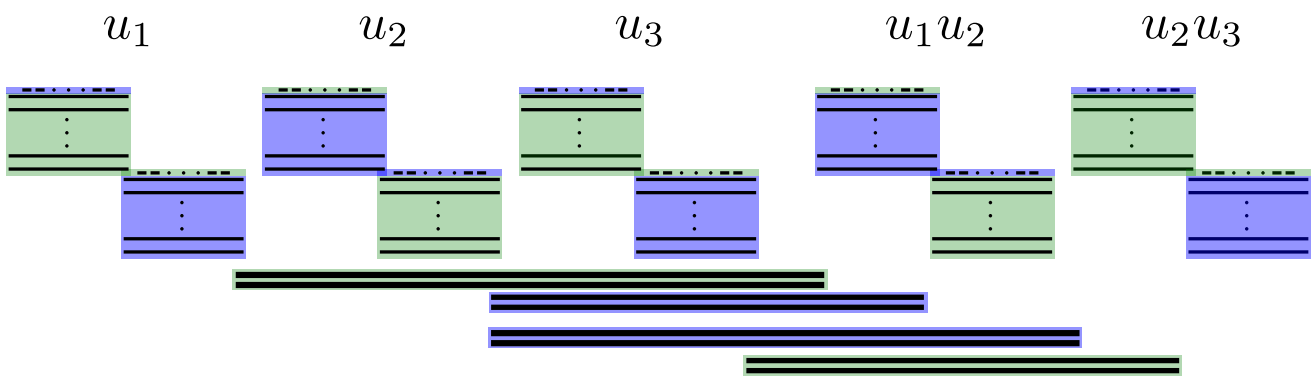


Adhikary, Bose, Mukherjee, and Roy's reduction

Polynomial-time reduction from MaxCut on cubic graphs.

Let G be a cubic graph, $\pi_V = (u_1, \ldots, u_n)$ and $\pi_E = (e_1, \ldots, e_m)$.

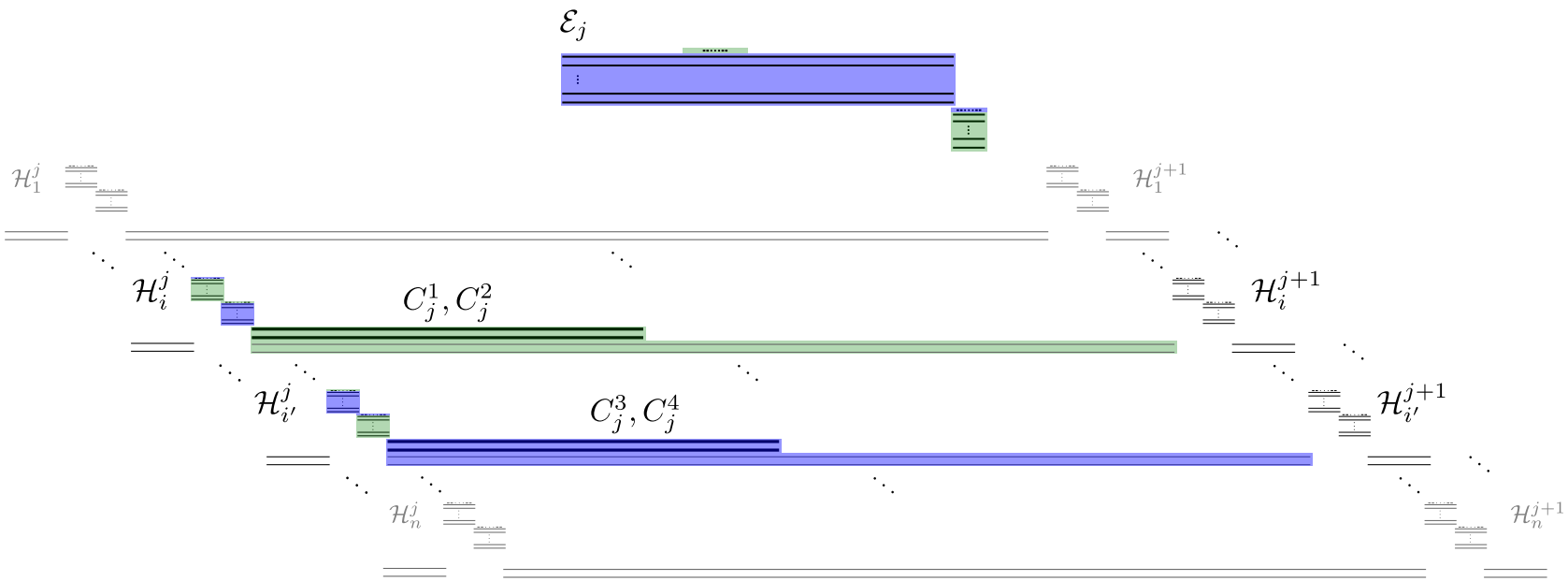
For suitable x, y, $\mathsf{mc}(\mathbb{G}_{\mathcal{M}}) \geq \phi(n, k)$ iff $\mathsf{mc}(G) \geq k$.



(Complexity of maximum cut on interval graphs. Adhikary, Bose, Mukherjee, Roy, 2021)

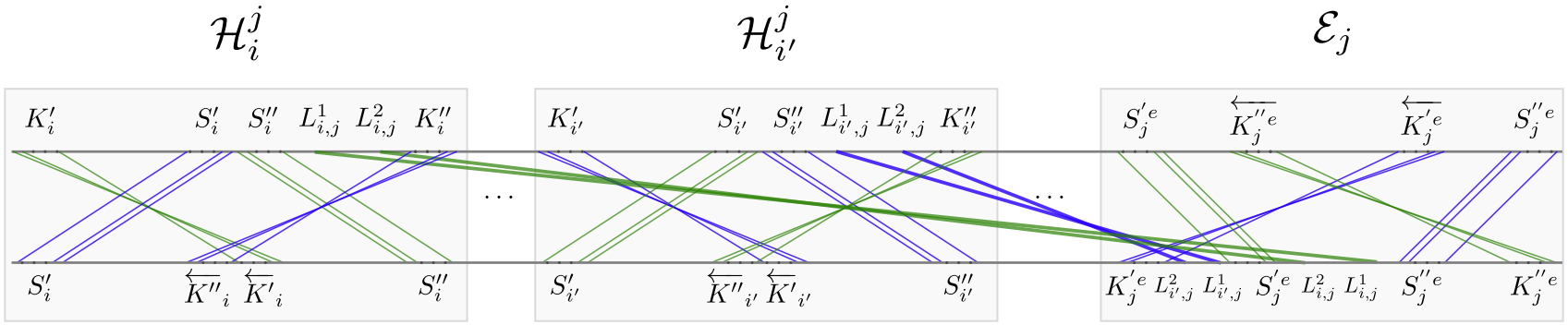
Our reduction: Interval count 4

$[A, B]$ is a max-cut of $\mathbb{G}_{\mathcal{M}}$



Our reduction: Permutation

$[A, B]$ is a max-cut of $\mathbb{G}_{\mathcal{M}}$



Main open questions

- Is MaxCut polynomial-time solvable on unit interval graphs?
- Is MaxCut polynomial-time solvable on interval permutation graphs?